BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

String Manipulation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Which CPU component performs arithmetic and logical operations?

2. Why might an HDD suit a large archive?
   A) No moving parts
   B) Low cost per GB
   C) Unlimited lifespan
   D) Always fastest access

3. "An SSD is faster because its magnetic read head rotates more quickly." Correct this. Explain why.

4. A laptop is used on a bumpy bus. "Portability and durability mean the same thing." Correct this. How should this affect the storage choice?

Today's objectives

Explain the core ideas of string manipulation using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Concatenation
Joining strings creates new text; it does not add numerical values.

Length
Counting characters helps validate usernames and input.

Slicing
A slice extracts a specified portion of a string, using the selected language's indexing rules.

Data type
Characters and strings represent text, even when they contain numeric-looking digits.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Concatenation
Joining strings creates new text; it does not add numerical values.

Length
Counting characters helps validate usernames and input.

Slicing
A slice extracts a specified portion of a string, using the selected language's indexing rules.

Data type
Characters and strings represent text, even when they contain numeric-looking digits.

I Do - Worked model

A school prints a username from firstName = 'Sam' and lastName = 'Lee'. Explain how to produce 'Sam.Lee'.

- Identify both strings and the separator.
- Concatenate firstName + '.' + lastName.
- The result is the string 'Sam.Lee'; do not add the text mathematically.
- Check that the period occurs once and no spaces were introduced.

We Do - Guided practice

1. A library creates an ID from 'BK' and '204'. What string is produced?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the OCR string operation to display NETWORK.

```
word = "network"
print(word....)
```

You Do - Independent exam questions

Q2. State the difference between concatenation and arithmetic addition. [2 marks]

Q3. A user enters 'AB' and '12'. Write the resulting concatenated string with no spaces. [1 mark]

Q4. Explain why checking string length can support validation of a password. [3 marks]

Exit Ticket - True or False?

Concatenating strings always adds their numeric values.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The ALU performs arithmetic and logical operations.

2. B - Low cost per GB

3. It does not need a moving head to seek positions on spinning platters.

4. Durability means resisting damage; portability means being easy to carry. An SSD has no moving read/write parts, so it handles shocks better. Size and weight also matter.

I Do - Explain the model

Identify both strings and the separator.

Concatenate firstName + '.' + lastName.

The result is the string 'Sam.Lee'; do not add the text mathematically.

Check that the period occurs once and no spaces were introduced.

We Do - Model answers

1. 'BK204'.

Code First - Answer

word = "network"
print(word.upper)
Expected output: NETWORK

You Do - Self-mark

Q2: Concatenation joins string content; arithmetic addition calculates a numerical sum.

Q3: AB12.

Q4: Length checks whether the input meets a stated minimum/maximum number of characters; alone it does not guarantee a secure password.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.